

Martingales

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Definition

A stochastic process in $\mathbb{R}$ is called a *martingale* with respect to $\{\mathcal{F}_t\}$ if

1. X_t is integrable for each $t \in T$,
2. $\{X_t\}$ is $\{\mathcal{F}_t\}$ adapted, and
3. for each $s, t \in T$ with $s < t$,

$$\mathbb{E}(X_t | \mathcal{F}_s) = X_s, \text{ a.s.} \quad (\star)$$

The process is called a $\mathcal{F}_t$ -submartingale if $(\star)$ is replaced by

$$\mathbb{E}(X_t | \mathcal{F}_s) \geq X_s \quad \text{a.s.}$$

The process is called a $\mathcal{F}_t$ -supermartingale if $(\star)$ is replaced by

$$\mathbb{E}(X_t | \mathcal{F}_s) \leq X_s \quad \text{a.s.}$$

When we say a process is a (sub-/super-) martingale without explicit mention of a filtration, then the underlying filtration is the natural/induced filtration.

Convex transformations of martingales

Lemma 1

(a) If X is a martingale, ϕ is convex and $\mathbb{E}(|\phi(X)|) < \infty$, then $\phi(X)$ is a submartingale.

(b) If X is a submartingale, ϕ is convex and increasing and $\mathbb{E}(|\phi(X)|) < \infty$, then $\phi(X)$ is a submartingale.

Proof.

Use Jensen's inequality.



Martingales transform I

In the sequel we assume that $T = \mathbb{Z}_+$ and we use n to denote the time parameter.

Example 2

Let $X_1, X_2, \dots$, be i.i.d. random variables with $\mathbb{E}X_1 = 0$. Then $S_n = \sum_{k=1}^n X_k$ is an $\mathcal{F}_n$ -martingale where $\mathcal{F}_n = \sigma\{X_1, \dots, X_n\}$.

A sequence of random variables $\{H_n : n \in \mathbb{Z}_+\}$ is said to be **predictable** if for each n , H_n is $\mathcal{F}_{n-1}$ measurable. Let $\{X_n\}$ be a martingale and $\{H_n\}$ be a bounded non-negative predictable process. Define a stochastic process $\{Y_n\}$ by

$$Y_0 = 0, \quad Y_n = Y_{n-1} + H_n \cdot (X_n - X_{n-1}), \quad n = 1, 2, \dots$$

that is, $Y_n = \sum_{k=1}^n H_k(X_k - X_{k-1})$. We then write $Y = H \cdot X$ and call it the martingale transform of X by H . (This is the discrete analog of Itô's stochastic integral).

Martingales transform II

Proposition 3

- (a) *If X is a martingale and H is predictable and bounded, then $Y = H \cdot X$ is a martingale.*
- (b) *If X is a submartingale and H is a non-negative bounded predictable process, then $Y = H \cdot X$ is a submartingale.*

Proof.

Direct verification of the defining properties.



Martingales transform III

Example 4 (Stopped Processes)

Let τ be a stopping time and let $H_n := 1_{\tau \geq n}$. Then H_n is predictable since

$$\{\tau \geq n\} = [\cup_{k=0}^{n-1} \{\tau = k\}]^c \in \mathcal{F}_{n-1}.$$

Then $X^\tau := H \cdot X$ is given by

$$X_n^\tau = X_{n \wedge \tau} \quad \text{for } n \in \mathbb{Z}_+.$$

Doob Decomposition

Lemma 5 (Doob Decomposition)

Any integrable and $\mathcal{F}$ -adapted process X on $\mathbb{Z}_+$ has an a.s. unique decomposition $M + A$, where M is an $\mathcal{F}$ -martingale and A is an $\mathcal{F}$ -predictable process with $A_0 = 0$. In particular, X is a submartingale iff A is a.s. nondecreasing.

Proof.

If $X = M + A$ for processes M and A as stated, then clearly $\Delta A_n = \mathbb{E}[\Delta X_n | \mathcal{F}_{n-1}]$, a.s., $\forall n \in \mathbb{Z}_+$, and so

$$A_n = \sum_{k \leq n} \mathbb{E}[\Delta X_k | \mathcal{F}_{k-1}] \text{ a.s., } n \in \mathbb{N}.$$

which proves the uniqueness. In general, we can define A as above. Then $M = X - A$ is clearly a martingale. □

Optional Sampling Theorem I

Theorem 6 (Doob's Optional Sampling Theorem)

Let $\{X_n : n \in \mathbb{Z}_+\}$ be a martingale (resp. submartingale/supermartingale with respect to $\{\mathcal{F}_n\}$ and let τ_1, τ_2 be bounded stopping times with $\tau_1 \leq \tau_2$. Then

- (a) $\mathbb{E}[X_{\tau_2} | \mathcal{F}_{\tau_1}] = X_{\tau_1}$ a.s. (resp. $\geq, \leq$). In particular*
- (b) $\mathbb{E}[X_{\tau_2}] = \mathbb{E}[X_{\tau_1}]$ (resp. $\geq, \leq$).*

Optional Sampling Theorem II

Proof.

(a) First note X_{τ_1} is $\mathcal{F}_{\tau_1}$ -measurable. Secondly, note that if a stopping time $\tau \leq u < \infty$ and if X is a martingale, then for any $A \in \mathcal{F}_\tau$,

$$\begin{aligned}\mathbb{E}[X_u; A] &= \sum_{k \leq u} \mathbb{E}[X_u; A \cap \{\tau = k\}] \\ &= \sum_{k \leq u} \mathbb{E}[X_k; A \cap \{\tau = k\}] \quad (\because \mathbb{E}[X_u | \mathcal{F}_k] = X_k, A \cap \{\tau = k\} \in \mathcal{F}_k) \\ &= \sum_{k \leq u} \mathbb{E}[X_\tau; A \cap \{\tau = k\}] = \mathbb{E}[X_\tau; A],\end{aligned}$$

and thus $\mathbb{E}[X_u | \mathcal{F}_\tau] = X_\tau$ a.s. Therefore, by $\tau_1 \leq \tau_2 \leq u$, we have $\mathcal{F}_{\tau_1} \subset \mathcal{F}_{\tau_2}$ and $\mathbb{E}[X_{\tau_2} | \mathcal{F}_{\tau_1}] = \mathbb{E}[\mathbb{E}[X_u | \mathcal{F}_{\tau_2}] | \mathcal{F}_{\tau_1}] = \mathbb{E}[X_u | \mathcal{F}_{\tau_1}]$. The sub- and super- martingale cases follow similarly.

(b) Direct implication of (a). □

Optional Sampling Theorem III

Corollary 7

Let $\{X_t\}$ satisfy (i) and (ii) in the definition of a martingale. Then it is a martingale (respectively, supermartingale, submartingale) if and only if

$$\mathbb{E}(X_{\tau_1}) = \mathbb{E}(X_{\tau_2}) \quad (\text{respectively } \leq, \geq)$$

for every pair of bounded stopping times $\tau_1 \leq \tau_2$ that take at most countable many different values.

Optional Sampling Theorem IV

Proof.

The only if part is by Theorem 6. For the if part, we show the martingale case only, the proof of the other cases being similar. It suffices to show $\mathbb{E}[X_t; A] = \mathbb{E}[X_s; A]$ for all $s \leq t$ and $A \in \mathcal{F}_s$. But for $s \leq t$, and $A \in \mathcal{F}_s$, $\tau = s1_A + t1_{A^c}$ and t are both stopping times and $\tau \leq t$, so

$$\begin{aligned}\mathbb{E}[X_t; A] - \mathbb{E}[X_s; A] &= \mathbb{E}[X_t] - \mathbb{E}[X_t; A^c] - \mathbb{E}[X_s; A] \\ &= \mathbb{E}[X_t] - \mathbb{E}[X_\tau] = 0\end{aligned}$$



Doob's Inequalities I

Theorem 8 (Doob's Maximal Inequalities)

Let $\{X_n : n \in \mathbb{Z}_+\}$ be a submartingale. Then $\forall N \in \mathbb{Z}_+, \forall \lambda > 0$,

$$(a) \quad \lambda P\left\{\max_{0 \leq n \leq N} X_n \geq \lambda\right\} \leq \mathbb{E}\left[X_N; \left\{\max_{0 \leq n \leq N} X_n \geq \lambda\right\}\right] \leq \mathbb{E}X_N^+.$$

$$(b) \quad \lambda P\left\{\min_{0 \leq n \leq N} X_n \leq -\lambda\right\} \leq -\mathbb{E}X_0 + \mathbb{E}[X_N^+] \leq \mathbb{E}|X_0| + \mathbb{E}|X_N|.$$

$$(c) \quad \lambda P\left\{\max_{0 \leq n \leq N} |X_n| \geq \lambda\right\} \leq \mathbb{E}|X_0| + 2\mathbb{E}|X_N|.$$

This result provides a means to control the large deviation probability of a martingale by its moment, and is an extension of the elementary Markov's (or Chebyshev's) inequality:

$$\lambda P(\xi \geq \lambda) \leq \mathbb{E}[\xi; \xi \geq \lambda] \leq \mathbb{E}[\xi^+] \leq \mathbb{E}[|\xi|].$$

Doob's Inequalities II

Proof.

(a) Let $\tau = \min\{n; X_n \geq \lambda\} \wedge N$, and $B = \{\max_{0 \leq n \leq N} X_n \geq \lambda\}$. Then τ is a stopping time $\leq N$, $B \in \mathcal{F}_\tau$, and $X_\tau \geq \lambda$ on B . Therefore,

$$\lambda P(B) \leq \mathbb{E}[X_\tau; B] \leq \mathbb{E}[X_N; B] \leq \mathbb{E}[X_N^+].$$

(b) Let $X = M + A$ be the Doob decomposition of X . Then $A \geq 0$ a.s., and by applying (a) on $-M$, we have

$$\begin{aligned} \lambda P\left(\min_{0 \leq n \leq N} X_n \leq -\lambda\right) &\leq \lambda P\left(\min_{0 \leq n \leq N} M_n \leq -\lambda\right) \leq \mathbb{E}[(-M)_N^+] \\ &= \mathbb{E}[M_N^-] = \mathbb{E}[M_N^+] - \mathbb{E}[M_N] \leq \mathbb{E}[X_N^+] - \mathbb{E}[X_0] \leq \mathbb{E}[|X_N|] + \mathbb{E}[|X_0|]. \end{aligned}$$

(c) Combine (a) and (b). □

Doob's Inequalities III

Corollary 9

Let $X = \{X_n\}$ be a martingale and $p \in [1, \infty)$. Then for every $N \in \mathbb{Z}_+$,

(d) $P\{\max_{0 \leq n \leq N} |X_n| \geq \lambda\} \leq \mathbb{E}[|X_N|^p] / \lambda^p;$

(e) and if $p > 1$, $\|\max_{0 \leq n \leq N} |X_n|\|_p \leq \frac{p}{p-1} \|X_N\|_p.$

Note part (d) implies that if a martingale satisfies stronger moment conditions, then its deviation probability will be “smaller” (tending to zero faster when the threshold value tends to infinity). Part (e) says that the p -th moment of the running maximum of the absolute value of a martingale up to time N can be controlled by the p -th moment of its absolute value at time N (up to a constant).

Doob's Inequalities IV

Proof.

(d) By the fact that $|X|^p$ is a submartingale.

(e) By Fubini's theorem, Doob's inequality, and Hölder's inequality,

$$\begin{aligned}\mathbb{E}[\max_{0 \leq n \leq N} |X_n|^p] &= p \mathbb{E} \left[\int_0^\infty 1_{\{\max_{0 \leq n \leq N} |X_n| \geq x\}} x^{p-1} dx \right] \\&= p \int_0^\infty P\{\max_{0 \leq n \leq N} |X_n| \geq x\} x^{p-1} dx \leq p \int_0^\infty \mathbb{E}[|X_N|; \max_{0 \leq n \leq N} |X_n| \geq x] x^{p-2} dx \\&= p \mathbb{E} \left[|X_N| \int_0^\infty 1_{\{\max_{0 \leq n \leq N} |X_n| \geq x\}} x^{p-2} dx \right] = \frac{p}{p-1} \mathbb{E} \left[|X_N| \max_{0 \leq n \leq N} |X_n|^{p-1} \right] \\&\leq p/(p-1) \|X_N\|_p \left\| \max_{0 \leq n \leq N} |X_n|^{p-1} \right\|_q \quad [\text{Here } q = p/(p-1)] \\&= p/(p-1) \|X_N\|_p \left\| \max_{0 \leq n \leq N} |X_n| \right\|_p^{p-1}.\end{aligned}$$

Now divide both sides by $\left\| \max_{0 \leq n \leq N} |X_n| \right\|_p^{p-1}$.

□

Doob's Inequalities V

Corollary 10

If X_n is a martingale, then $\mathbb{E}[\max_{0 \leq n \leq N} |X_n|^2] \leq 4\mathbb{E}|X_N|^2$.

Proof.

Set $p = 2$ in Doob's inequality (e).



Upcrossing Inequality I

Let $\{X_n\}$ be an $\{\mathcal{F}_n\}$ -adapted process. For any $a < b$, let $U_a^b(t)$ denote the number of $[a, b]$ -crossings of X up to time t , defined as the supremum of all $n \in \mathbb{Z}_+$ such that there exist times $s_1 < t_1 < s_2 < t_2 < \cdots < s_n < t_n \leq t$ with $X_{s_k} \leq a$ and $X_{t_k} \geq b$ for all k .

Theorem 11 (Upcrossing inequality)

Let $X = \{X_n : n \in \mathbb{Z}_+\}$ be a submartingale. Then for every $n \in \mathbb{Z}_+$ and real numbers $a < b$,

$$\mathbb{E}[U_a^b(n)] \leq \frac{1}{b-a} \mathbb{E}[(X_n - a)^+].$$

Upcrossing Inequality II

Proof.

Since $Y = (X - a)^+$ is again a submartingale and the $[a, b]$ -crossings of X correspond to the $[0, b - a]$ -crossings of Y , we can assume $X \geq 0$ and $a = 0$. Now define recursively the optional times

$0 = \tau_0 \leq \sigma_1 < \tau_1 < \sigma_2 < \dots$ by

$$\sigma_k = \inf\{n \geq \tau_{k-1}; X_n = 0\}, \quad \tau_k = \inf\{n \geq \sigma_k; X_n \geq b\}, \quad k \in \mathbb{N},$$

and introduce the predictable process $V_n = \sum_{k \geq 1} 1\{\sigma_k < n \leq \tau_k\}$, $n \in \mathbb{N}$. Then the martingale transform $(1 - V) \cdot X$ is again a submartingale, and so

$$\mathbb{E}[(1 - V) \cdot X)_n] \geq \mathbb{E}[(1 - V) \cdot X)_0] = 0, \quad n \geq 0.$$

Since also $(V \cdot X)_n \geq bU_0^b(n)$, we have

$$b\mathbb{E}[U_0^b(n)] \leq \mathbb{E}[(V \cdot X)_n] \leq \mathbb{E}[(1 \cdot X)_n] = \mathbb{E}[X_n] - \mathbb{E}[X_0] \leq \mathbb{E}[X_n].$$



Martingale convergence theorem

Theorem 12 (Forward Submartingale Convergence Theorem)

Let $X_1, X_2, \dots$ be a submartingale such that $\sup_n \mathbb{E}[X_n^+] < \infty$.

Then

- (i) $X_\infty = \lim_{n \rightarrow \infty} X_n$, exists a.s..
- (ii) $\mathbb{E}|X_\infty| < \infty$.

Proof.

First note

$$\sup_n \mathbb{E}|X_n| = \sup_n (2\mathbb{E}[X_n^+] - \mathbb{E}[X_n]) \leq \sup_n (2\mathbb{E}[X_n^+] - \mathbb{E}[X_1]) < \infty.$$

(i) By the Doob inequalities, $\sup_n |X_n| < \infty$ a.s.. By the upcrossing inequality, X has a.s. finitely many upcrossings of every interval $[a, b]$ with rational $a < b$. Outside the null set where any of these conditions fails, $X_\infty = \lim_n X_n$ clearly exists.

(ii) By Fatou's Lemma,

$$\mathbb{E}|X_\infty| = \mathbb{E} \liminf_n |X_n| \leq \liminf_n \mathbb{E}|X_n| \leq \sup_n \mathbb{E}|X_n| < \infty. \quad \square$$

Uniform integrability and closability of martingales I

A family $\{Y_\alpha\}$ of random variables is said to be **uniformly integrable** if

$$\lim_{K \rightarrow \infty} \sup_{\alpha} \mathbb{E}[|Y_\alpha|; |Y_\alpha| > K] = 0.$$

Also recall that for $p \in (0, \infty)$ if $\|X\|_p < \infty$ we say $X \in L^p$ and if $X_n, n = 1, 2, \dots, X \in L^p$ and $\|X_n - X\|_p \rightarrow 0$ ($n \rightarrow \infty$) we say $X_n \rightarrow X$ in L^p .

Uniform integrability and closability of martingales II

Lemma 13

Assume $\mathbb{E}|X| < \infty$. Given $\epsilon > 0$ there exists $\delta > 0$ such that if $F \in \mathcal{F}$, $P(F) \leq \delta$, then $\mathbb{E}[|X|; F] < \epsilon$

Proof.

By dominated convergence,

$\mathbb{E}[|X|; |X| > n] = \mathbb{E}|X| - \mathbb{E}[|X|; |X| \leq n] \downarrow 0$, so for any $\epsilon > 0$, there exists N_ϵ s.t. $\mathbb{E}[|X|; |X| \geq N_\epsilon] < \epsilon/2$. Let $\delta = \epsilon/(2N_\epsilon)$. Then if $P(F) \leq \delta$, we have

$$\begin{aligned}\mathbb{E}[|X|; F] &= \mathbb{E}[|X|; F \cap \{|X| \leq N_\epsilon\}] + \mathbb{E}[|X|; F \cap \{|X| > N_\epsilon\}] \\ &< N_\epsilon P(F) + \epsilon/2 \leq \epsilon.\end{aligned}$$

□

Uniform integrability and closability of martingales III

Lemma 14

The family of random variables $\{X_\alpha\}$ is uniformly integrable if and only if

- (i) (L^1 boundedness) $\sup_\alpha \mathbb{E}|X_\alpha| < \infty$;
- (ii) (uniform continuity in probability) $\lim_{PA \rightarrow 0} \sup_\alpha \mathbb{E}[|X|_\alpha; A] = 0$.

Proof.

Conside the “only if” part first. Note

$\mathbb{E}[|X_\alpha|; A] \leq KP(A) + \mathbb{E}[|X_\alpha|; |X_\alpha| > K]$, for all $K > 0$, A measurable, and any α . For any $\epsilon > 0$, take a K_0 such that $\mathbb{E}[|X_\alpha|; |X_\alpha| > K_0] < \epsilon/2$ and $\delta = \epsilon/(2K_0)$. Then $PA < \delta$ implies $\mathbb{E}[|X_\alpha|; A] < \epsilon$ and (ii) follows. Also, (i) follows by taking $A = \Omega$ and noting $\mathbb{E}|X_\alpha| \leq K_0 + \epsilon/2 < \infty$.

Now assume both (i) and (ii). Note

$\sup_\alpha P\{|X_\alpha| > K\} \leq \sup_\alpha \mathbb{E}|X_\alpha|/K \rightarrow 0$ as $K \rightarrow \infty$. Therefore $\sup_\alpha \mathbb{E}[|X_\alpha|; |X_\alpha| > K] \rightarrow 0$ follows by (ii) with $A = \{|X_\alpha| > K\}$. □

Lemma 15

If $X_n \rightarrow X$ a.s. and X_n is uniformly integrable, then $\|X_n\|_1 \rightarrow \|X\|_1$ (i.e. $\mathbb{E}|X_n| \rightarrow \mathbb{E}|X|$) and $X_n \rightarrow X$ in L^1 (i.e. $\mathbb{E}|X_n - X| \rightarrow 0$).

Uniform integrability and closability of martingales IV

Proof.

First note for any $K > 0$,

$$\begin{aligned} |\mathbb{E}|X_n| - \mathbb{E}|X|| &\leq |\mathbb{E}[|X_n| - |X_n| \wedge K]| + |\mathbb{E}[|X_n| \wedge K - |X| \wedge K]| \\ &\quad + |\mathbb{E}[|X| \wedge K - |X||] \\ &\leq \sup_n \mathbb{E}[|X_n|; |X_n| > K] + \mathbb{E}[|X_n| \wedge K - |X| \wedge K] \\ &\quad + \mathbb{E}[|X|; |X| > K]. \end{aligned}$$

By dominated convergence

$$\limsup_n |\mathbb{E}|X_n| - \mathbb{E}|X|| \leq \sup_n \mathbb{E}[|X_n|; |X_n| > K] + \mathbb{E}[|X|; |X| > K]$$

for any $K > 0$. Letting $K \rightarrow \infty$ implies $\limsup_n |\mathbb{E}|X_n| - \mathbb{E}|X|| = 0$ or equivalently $\mathbb{E}|X_n| \rightarrow \mathbb{E}|X|$.

Now note that $|X_n - X| \leq |X_n| + |X|$, that $|X_n| + |X| \rightarrow 2|X|$ a.s. and that $\mathbb{E}[|X_n| + |X|] \rightarrow 2\mathbb{E}|X| < \infty$, and so by dominated convergence, $\lim_n \mathbb{E}|X_n - X| = \mathbb{E}_n \lim |X_n - X| = \mathbb{E}[0] = 0$. □

Uniform integrability and closability of martingales V

Remark 16

By the Minkowski inequality we have $|\mathbb{E}|X_n| - \mathbb{E}|X|| \leq \mathbb{E}|X_n - X|$ therefore under the conditions of the lemma, we actually have $\|X_n\|_1 \rightarrow \|X\|_1 \iff X_n \rightarrow X$ in L^1 . In fact, together with the almost sure convergence condition, either of these two convergences also implies the uniform integrability of the sequence. [To see this, assume $\|X_n\|_1 \rightarrow \|X\|_1$ or equivalently $\mathbb{E}|X_n| \rightarrow \mathbb{E}|X|$. First note that for any fixed $K > 0$, by the continuity of the function $x \wedge (K - x)^+$ and dominated convergence

$$\mathbb{E}[|X_n|; |X_n| > K] \leq \mathbb{E}[|X_n| - |X_n| \wedge (K - |X_n|)^+] \rightarrow \mathbb{E}[|X| - |X| \wedge (K - |X|)^+]$$

Now, since $|X| \wedge (K - |X|)^+ \uparrow |X|$ as $K \rightarrow \infty$, the RHS tends to 0 by dominated convergence. Therefore, the LHS can be arbitrarily small when $K \rightarrow \infty$.]

In summary, we have the stronger result: If $X_1, X_2, \dots, X \in L^1$ and $X_n \rightarrow X$ a.s., then the three conditions, (1) $\{X_n\}$ is uniformly integrable; (2) $\|X_n\|_1 \rightarrow \|X\|_1$; and (3) $X_n \rightarrow X$ in L^1 are equivalent.

Uniform integrability and closability of martingales VI

Theorem 17

Assume $\mathbb{E}|X| < \infty$. Then the collection of random variables $\{\mathbb{E}[X|\mathcal{G}]; \mathcal{G} \text{ is a sub-}\sigma\text{-field of } \mathcal{F}\}$ is uniformly integrable.

Proof.

First note

$$\mathbb{E}[|\mathbb{E}^{\mathcal{G}}X|; |\mathbb{E}^{\mathcal{G}}X| > K] \leq \mathbb{E}[\mathbb{E}^{\mathcal{G}}|X|; |\mathbb{E}^{\mathcal{G}}X| > K] = \mathbb{E}[|X|; |\mathbb{E}^{\mathcal{G}}X| > K]$$

Then note $P\{|\mathbb{E}^{\mathcal{G}}X| > K\} \leq \mathbb{E}|\mathbb{E}^{\mathcal{G}}X|/K \leq \mathbb{E}|X|/K \rightarrow 0$ as $K \rightarrow \infty$ and use Lemma 13 to see $\mathbb{E}[|X|; |\mathbb{E}^{\mathcal{G}}X| > K] \rightarrow 0$ as $K \rightarrow \infty$. $\square$

Uniform integrability and closability of martingales VII

A martingale/submartingale/supermartingale X on $\mathbb{Z}_+$ is said to be **closable** if there exists a $\xi \in L^1$ (i.e. integrable) such that $\{X_n; n \in \overline{\mathbb{Z}}_+\}$ with $X_\infty \triangleq \xi$ is a martingale/submartingale/supermartingale relative to the filtration $\{\mathcal{F}_n; n \in \overline{\mathbb{Z}}_+\}$, where $\overline{\mathbb{Z}}_+ = \mathbb{Z}_+ \cup \{\infty\}$ and $\mathcal{F}_\infty \triangleq \sigma(\cup_n \mathcal{F}_n)$.

Theorem 18

Let $\{X_n; n \in \mathbb{Z}_+\}$ be a submartingale. Then X is closable if and only if $\{X_n^+, n \in \mathbb{Z}_+\}$ is uniformly integrable.

Uniform integrability and closability of martingales VIII

Proof.

The “only if” part: If there is an $X_\infty \in L^1$ such that X_n , $n \in \overline{\mathbb{Z}}_+$ is a submartingale, then X_n^+ , $n \in \overline{\mathbb{Z}}_+$ is a submartingale too [by convexity of $(\cdot)^+$ and definition of integrability]. Therefore, $\sup_{n \in \mathbb{Z}_+} \mathbb{E}[X_n^+; X_n^+ > K] \leq \sup_{n \in \overline{\mathbb{Z}}_+} \mathbb{E}[X_\infty^+; X_n^+ > K] \rightarrow 0$ as $K \rightarrow \infty$ (by Lemma 13 and $\sup_n P\{X_n^+ > K\} \leq \sup_n \mathbb{E}X_n^+/K \rightarrow 0$ as $K \rightarrow \infty$). To show the “if” part, assume the uniform integrability condition on X_n^+ . First note $\sup_{n \in \mathbb{Z}_+} \mathbb{E}[X_n^+] < \infty$ by Lemma 14, and therefore $X_\infty \triangleq \lim X_n$ exists and $\mathbb{E}|X_\infty| < \infty$ by Theorem 12. By Lemma 15, $X_n^+ \rightarrow X_\infty^+$ in L^1 . Thus, by the contractivity of conditional expectations, we have for all m , $\mathbb{E}[X_n^+|\mathcal{F}_m] \rightarrow \mathbb{E}[X_\infty^+|\mathcal{F}_m]$ in L^1 and thus in probability as well, which in turn implies that the convergence is a.s. along a subsequence n_k , $k = 1, 2, \dots$. By the submartingale property, we have $X_m \leq \mathbb{E}[X_{n_k}|\mathcal{F}_m]$ for all k large, and so, a.s.,

$$\begin{aligned} X_m &\leq \liminf_k \mathbb{E}[X_{n_k}|\mathcal{F}_m] = \liminf_k (\mathbb{E}[X_{n_k}^+|\mathcal{F}_m] - \mathbb{E}[X_{n_k}^-|\mathcal{F}_m]) \\ &= \lim_k \mathbb{E}[X_{n_k}^+|\mathcal{F}_m] - \liminf_k \mathbb{E}[X_{n_k}^-|\mathcal{F}_m] \\ &\leq \mathbb{E}[X_\infty^+|\mathcal{F}_m] - \mathbb{E}[\liminf_k X_{n_k}^-|\mathcal{F}_m] \quad (\text{By Fatou's Lemma}) \\ &= \mathbb{E}[X_\infty|\mathcal{F}_m]. \end{aligned}$$

□

Uniform integrability and closability of martingales IX

Remark 19

In the proof we have used the following general results.

(1) If $\xi_n \rightarrow \xi$ in L^1 then $\xi_n \rightarrow \xi$ in probability [because for all $\epsilon > 0$,

$$P(|\xi_n - \xi| > \epsilon) \leq \mathbb{E}|\xi_n - \xi|/\epsilon \rightarrow 0].$$

(2) If $\xi_n \rightarrow \xi$ in probability, then $\mathbb{E}[|\xi_n - \xi| \wedge 1] \rightarrow 0$ [because for arbitrarily small $\epsilon > 0$,

$$\begin{aligned}\mathbb{E}[|\xi_n - \xi| \wedge 1] &= \mathbb{E}[|\xi_n - \xi| \wedge 1; |\xi_n - \xi| \leq \epsilon] + \mathbb{E}[|\xi_n - \xi| \wedge 1; |\xi_n - \xi| > \epsilon] \\ &\leq \epsilon + P(|\xi_n - \xi| > \epsilon),\end{aligned}$$

which implies $\limsup_n \mathbb{E}[|\xi_n - \xi| \wedge 1] \leq \epsilon$ for all $\epsilon > 0$ and thus

$$\limsup_n \mathbb{E}[|\xi_n - \xi| \wedge 1] = 0.]$$

(3) If $\xi_n \rightarrow \xi$ in probability then there exists a subsequence $\{\xi_{n_k}\}_{k=1}^\infty$ along which the convergence is almost sure. [By (2), $\mathbb{E}[|\xi_n - \xi| \wedge 1] \rightarrow 0$, so we can find an increasing sequence $\{n_k\}$ such that $\mathbb{E}[|\xi_{n_k} - \xi| \wedge 1] \leq 2^{-k}$ for all k .

Then $\mathbb{E}[\sum_{k=1}^\infty |\xi_{n_k} - \xi| \wedge 1] = \sum_{k=1}^\infty \mathbb{E}[|\xi_{n_k} - \xi| \wedge 1] < \sum_k 2^{-k} < \infty$, and so almost surely $\sum_{k=1}^\infty |\xi_{n_k} - \xi| \wedge 1 < \infty$ and $|\xi_{n_k} - \xi| \wedge 1 \rightarrow 0$, which is the same as $\xi_{n_k} \rightarrow \xi$.]

Reverse Submartingale Convergence

Theorem 20 (Reverse Submartingale Convergence Theorem)

Let $\{X_n, n = 0, -1, -2, -3, \dots\}$ be a submartingale, then $\lim_{n \rightarrow -\infty} X_n = X_{-\infty}$ exists a.s. and $-\infty \leq X_{-\infty} < \infty$. Moreover, the following conditions are equivalent:

- (a) $\{X_n\}$ is uniformly integrable.
- (b) $X_n \rightarrow X_{-\infty}$ in L^1 .
- (c) $\{X_n; n = 0, -1, \dots, -\infty\}$ is a submartingale (with $\mathcal{F}_{-\infty} = \cap_{n \leq 0} \mathcal{F}_n$).
- (d) $\inf_{-\infty < n \leq 0} \mathbb{E}[X_n] > -\infty$.

Proof.

Reverse Submartingale Convergence

Proof.

Let $U_n^X(a, b)$ be the number of upcrossings of $[a, b]$ by $X_{-n}, \dots, X_0$ and let $U_\infty^X(a, b) = \lim_n U_n^X(a, b)$. As in the proof of Theorem 12, by $\mathbb{E}[U_n^X(a, b)] \leq (\mathbb{E}[X_0^+] + |a|)/(b - a)$ for all n , we have $\mathbb{E}[U_\infty^X(a, b)] < \infty$ and therefore $U_\infty^X(a, b) < \infty$ a.s. for all rational $a < b$, which in turn implies

$P\{\liminf_{n \rightarrow -\infty} X_n < \limsup_{n \rightarrow -\infty} X_n\} = 0$. Therefore $\liminf X_n = \limsup X_n = \lim X_n = X_{-\infty}$ exists a.s. Moreover, by $\mathbb{E}[X_{-\infty}^+] \leq \liminf \mathbb{E}X_n^+ \leq \mathbb{E}X_0^+ < \infty$ we also have $X_{-\infty} \in [-\infty, \infty)$ a.s.

Reverse Submartingale Convergence

Proof.

(a) $\Rightarrow$ (b) is by Lemma 15. To show (b) $\Rightarrow$ (c), we note for any $-m \leq -n$ and $A \in \mathcal{F}_{-\infty} \subset \mathcal{F}_{-m} \subset \mathcal{F}_{-n}$, by the submartingale property we have $\mathbb{E}[X_{-m}1_A] \leq \mathbb{E}[X_{-n}1_A]$. Since $X_{-m} \rightarrow X_{-\infty}$ ($m \rightarrow \infty$) in L^1 also implies $X_{-m}1_A \rightarrow X_{-\infty}1_A$ in L^1 and thus $\mathbb{E}[X_{-m}1_A] \rightarrow \mathbb{E}[X_{-\infty}1_A]$, letting $m \rightarrow \infty$ we have $\mathbb{E}[X_{-\infty}1_A] \leq \mathbb{E}[X_{-n}1_A]$, which implies $X_{-\infty} \leq \mathbb{E}[X_{-n}|\mathcal{F}_{-\infty}]$. Since $X_{-\infty} = \lim_n X_{-n-m}$ a.s. is $\mathcal{F}_{-m}$ -measurable for all $-m$, it's also measurable with respect to $\bigcap_{m=1}^{\infty} \mathcal{F}_{-m} = \mathcal{F}_{-\infty}$. Therefore X_n ; $n \in \overline{\mathbb{Z}}_-$ is a submartingale. The implication (c) $\Rightarrow$ (d) is obvious since $\mathbb{E}[X_{-\infty}] > -\infty$.

Reverse Submartingale Convergence

Proof.

To show $(d) \Rightarrow (a)$, first note

$P(|X_n| > K) \leq \mathbb{E}|X_n|/K \leq (2\mathbb{E}X_0^+ - \inf \mathbb{E}[X_n])/K$ converges to 0 uniformly in n as $K \rightarrow \infty$. Therefore

$\mathbb{E}[X_n^+; X_n^+ \geq K] \leq \mathbb{E}[X_0^+; X_n^+ \geq K] \rightarrow 0$, as $K \rightarrow \infty$ (by Lemma 13). This shows that $\{X_n^+\}$ is uniformly integrable. Now note for $n < m \leq 0$,

$$\begin{aligned} 0 &\geq \mathbb{E}[X_n; X_n < -K] \geq \mathbb{E}[X_n] - \mathbb{E}[X_m; X_n \geq -K] \\ &= \mathbb{E}[X_n - X_m] + \mathbb{E}[X_m; X_n < -K], \end{aligned}$$

Since by (d) we can choose $-m$ big enough to make

$\mathbb{E}[X_n - X_m] > -\epsilon$, for such an m , we can choose K big enough to make $\mathbb{E}[|X_m|; X_n < -K] < \epsilon$ uniform in n . This implies X_n^- is also uniformly integrable. Thus X_n is uniformly integrable. $\square$

Some Applications I

Theorem 20 (Radon Nikodym Theorem)

Let $(\Omega, \mathcal{F}, P)$ be a countably generated probability space. That is, there exists a countable collection $\mathcal{C} \subset 2^\Omega$ such that $\mathcal{F} = \sigma(\mathcal{C})$. Suppose that Q is a finite measure on $(\Omega, \mathcal{F})$ which is absolutely continuous relative to P , that is, $P(F) = 0 \Rightarrow Q(F) = 0$. Then there exists X in $L^1(\Omega, \mathcal{F}, P)$ such that $Q = X \cdot P$, that is,

$$Q(F) = \int_F X(\omega) P(d\omega) \quad \forall F \in \mathcal{F}.$$

The random variable X is called the Radon-Nikodym derivative of Q with respect to P and is written as $X = \frac{dQ}{dP}$.

Proof.

Some Applications II

Let $\mathcal{C} = \{B_1, B_2, \dots\}$ be such that $\sigma(\mathcal{C}) = \mathcal{F}$, and $\mathcal{F}_n = \sigma(\{B_1, \dots, B_n\})$. Then $\mathcal{F}_n$ is a finite collection. Let $\mathcal{Z}_n$ be the collection of atomic sets of $\mathcal{F}_n$ ($B \in \mathcal{Z}_n$ means $B \neq \emptyset$ and the only $\mathcal{F}_n$ -measurable subsets of B are $\emptyset$ and B) and define

$$X_n = \sum_{C \in \mathcal{Z}_n} \frac{Q(C)}{P(C)} 1_C \quad (\text{define } \frac{0}{0} = 0 \text{ as usual}).$$

Then X_n can be verified to be a martingale with the property that $\mathbb{E}[X_n^+] = Q(\Omega) < \infty$ [In fact, X_n is the density of $Q|_{\mathcal{F}_n}$ with respect to $P|_{\mathcal{F}_n}$ and $\mathbb{E}[X_n; A] = Q(A)$ for all $A \in \mathcal{F}_n$]. Therefore X_n converges to an $X_\infty \in L^1$ a.s. by the martingale convergence theorem.

Now note that $Q \ll P$ implies $\forall \epsilon > 0, \exists \delta > 0$ s.t.: $P(A) < \delta \Rightarrow Q(A) < \epsilon$ (since otherwise $\exists \epsilon_0 > 0$ s.t.: $\forall n, \exists A_n, P(A_n) < 2^{-n}$ but $Q(A_n) \geq \epsilon_0 > 0$, which means $P(A_n \text{ i.o.}) = P(\bigcap_n \bigcup_{m \geq n} A_m) = 0$, but by Fatou's lemma,

$Q(\bigcap_n \bigcup_{m \geq n} A_m) \geq \limsup_n Q(A_n) \geq \epsilon_0 > 0$!). Therefore, for $\epsilon > 0$, when $K > Q(\Omega)/\delta$, then we have

$$P(X_n > K) \leq \mathbb{E}[X_n]/K = Q(\Omega)/K < \delta$$

and so $\mathbb{E}[X_n; X_n > K] = Q(X_n > K) < \epsilon$. Therefore $\{X_n\}$ is uniformly integrable and thus we also have $X_n \rightarrow X_\infty$ in L^1 , which in turn implies $X_n 1_A \rightarrow X_\infty 1_A$ in L^1 for all $A \in \mathcal{F}$. Therefore, $\mathbb{E}[X_\infty; F] = \lim_n \mathbb{E}[X_n; F] = Q(F)$ for all $F \in \mathcal{F}$. $\square$

Theorem 21 (Kakutani's Theorem on Product Martingales)

Let $X_1, X_2, \dots$ be independent non-negative random variables with mean 1. Define $M_0 \triangleq 1$ and for $n \in \mathbb{N}$,

$$M_n \triangleq X_1 X_2 \cdots X_n.$$

Then M is a non-negative martingale so that

$$M_\infty = \lim M_n \quad \text{exists a.s..}$$

The following five statements are equivalent:

- (i) $\mathbb{E}(M_\infty) = 1$.
- (ii) $M_n \rightarrow M_\infty$ in $\mathcal{L}^1$.
- (iii) M is uniformly integrable.
- (iv) $\prod a_n > 0$ or equivalently $\sum(1 - a_n) < \infty$, where $0 < a_n \triangleq \mathbb{E}(X_n^{1/2}) \leq 1$.

If any of these fails, then $P(M_\infty = 0) = 1$.

Proof.

Since $\sup_n \mathbb{E}[M_n^+] = \sup_n \mathbb{E}[M_n] = 1 < \infty$, we have $M_\infty = \lim_n M_n$ exists a.s. and $\in L^1$. Therefore, the equivalence of (i)-(iii) is due to Remark 16.

(iv) $\Rightarrow$ (iii): Let $N_n = \prod_{i=1}^n X_i^{1/2}/a_i$. Then

$\mathbb{E}N_n^2 = 1/(\prod_{i \leq n} a_i)^2 < 1/(\prod_{i=1}^\infty a_i)^2 < \infty$, and therefore by Doob's inequality $\mathbb{E} \sup_n M_n \leq \mathbb{E}[\sup_n N_n^2] \leq 4 \sup_n \mathbb{E}N_n^2 < \infty$. Now (iii) follows since $\sup_n M_n$ dominates the sequence M_n and is integrable.

(i) $\Rightarrow$ (iv): If $\prod a_n = 0$ then $M_n^{1/2} \rightarrow 0$ a.s. (since N_n is a positive martingale and so converges!). Then $M_n \rightarrow 0 = M_\infty$ a.s. and (i) cannot hold! □

Consistency of likelihood ratio test I

Let $f(x)$ and $g(x)$ be two density functions which are strictly positive. We observe iid random variables $X_1, X_2, \dots$ distributed with density function f or g (relative to some reference measure, say ν) and we wish to decide which on the basis of the observations.

Let P and Q be the measures on $\mathbb{R}^\infty$ which corresponds to iid random variables $\{X_1, X_2, \dots\}$ having density functions f, g respectively.

Consider the hypotheses:

$$\begin{aligned}H_f &\triangleq \{X'\text{'s have density } f\} \\H_g &\triangleq \{X'\text{'s have density } g\}.\end{aligned}$$

Let $r(x) \triangleq g(x)/f(x)$ and let $Y_n \triangleq r(X_n)$ and let $M_n \triangleq Y_1 Y_2 \cdots Y_n$ (likelihood ratio). We will reject H_g if the likelihood ratio is small. This is the likelihood ratio test. The following result assures us that under H_f when sample size $n \rightarrow \infty$ the likelihood ratio tends to 0 and therefore H_g is always rejected. That is, the likelihood ratio test is (strongly) consistent.

Consistency of likelihood ratio test II

Theorem 22 (Consistency of the Likelihood Ratio Test)

If the random variables are actually distributed according to P and $Q \neq P$, then $M_n \rightarrow 0$, P - a.s. as $n \rightarrow \infty$.

Proof.

Note $\sum_{i=1}^{\infty} (1 - \mathbb{E}_P \sqrt{g(X_i)/f(X_i)}) = \infty \cdot \frac{1}{2} \int (\sqrt{f(x)} - \sqrt{g(x)})^2 d\nu(x) = \infty$, and then use the previous result. □

The Strong Law of Large Number I

Theorem 23 (The Strong Law of Large Numbers)

Let $X_1, X_2, \dots$ be i.i.d. random variables with mean $\mathbb{E}X_1 = \mu \in (-\infty, \infty)$, and $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then $\bar{X}_n \rightarrow \mu$ a.s.

The Strong Law of Large Number II

Proof.

Let $Y_{-n} = \bar{X}_n$, $\mathcal{F}_{-n} = \sigma\{\bar{X}_n, \bar{X}_{n+1}, \dots\} = \sigma\{Y_k; k \leq -n\}$, $n = 1, 2, \dots$. Then Y_n , $n \in -\mathbb{N}$ is a martingale with respect to the filtration $\mathcal{F}_n$, $n \in -\mathbb{N}$. To see this, it suffices to verify $\mathbb{E}[Y_{-n} | \mathcal{F}_{-n-1}] = Y_{-n-1}$. But $\mathbb{E}[X_i | \sum_{i=1}^{n+1} X_i] \equiv \mathbb{E}[X_1 | \sum_{i=1}^{n+1} X_i] = \frac{1}{n+1} \sum_{i=1}^{n+1} X_i = \bar{X}_{n+1}$, $i = 1, \dots, n+1$ by symmetry, and therefore

$$\begin{aligned}\mathbb{E}[Y_{-n} | \mathcal{F}_{-n-1}] &= \mathbb{E}[\bar{X}_n | \sigma\{\bar{X}_{n+1}, \bar{X}_{n+2}, \dots\}] \\ &= \mathbb{E}[\bar{X}_n | \sum_{i=1}^{n+1} X_i, X_{n+2}, \dots] = \mathbb{E}[\bar{X}_n | \sum_{i=1}^{n+1} X_i] = \bar{X}_{n+1} = Y_{-n-1}.\end{aligned}$$

By the reverse submartingale convergence theorem, $\bar{X}_n = Y_{-n} \rightarrow Y_{-\infty}$ a.s. and $\{Y_n; n \in -\mathbb{N}\}$ is a martingale. So $\mathbb{E}[Y_{-\infty}] = \mu$.

It remains to show $Y_{-\infty}$ is constant a.s. However, $Y_{-\infty} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i$ is measurable with respect to $\mathcal{T} = \bigcap_{n=1}^{\infty} \sigma\{X_m; m \geq n\}$. Note that for all k , $\mathcal{T} = \bigcap_{n=k+1}^{\infty} \sigma\{X_m; m \geq n\}$ and therefore $\mathcal{T}$ is independent of $X_1, X_2, \dots, X_k$ and so also independent of $\sigma\{X_1, X_2, \dots\} \supset \mathcal{T}$. This means any $\mathcal{T}$ -measurable event is independent of itself and so must have probability 0 or 1. Therefore, $Y_{-\infty}$ as a $\mathcal{T}$ -measurable random variable has to equal a constant with probability 1. This constant must be μ since $\mathbb{E}[Y_{-\infty}] = \mu$.

